

Jacob W. Toney

jwt@mit.edu | jwtony.github.io | Google Scholar | linkedin.com/in/jacob-w-tony

RESEARCH PROFILE

Ph.D. candidate in Chemical Engineering and Computational Science at MIT working at the intersection of machine learning and quantum chemistry. Developing geometric deep learning models and large quantum chemical datasets to predict the structural, electronic, and chemical properties of transition metal complexes. Additional technical experience in quantum computing algorithms for chemistry (IBM Research) and large-scale operations engineering (SpaceX).

EDUCATION

Massachusetts Institute of Technology

Ph.D., *Chemical Engineering and Computational Science* (Minor: *Quantum Chemistry*)

Cambridge, MA

Sep. 2022 – Jul. 2027

Doctoral Advisor: Prof. Heather J. Kulik

Thesis: "Accelerating Quantum Chemistry with Machine Learning for the Rational Design of Transition Metal Complexes"

Selected Coursework: Electronic Structure Theory, Computational Chemistry, Machine Learning, Bayesian Modeling and Inference, Diffusion Models, Generative AI and Stochastic Differential Equations, Matrix Calculus, Numerical Methods

University of Southern California

M.S., *Chemical Engineering*

B.S., *Chemical Engineering* (Minor: *Materials Science*) | GPA: 4.00

Los Angeles, CA

Sep. 2019 – May 2021

Aug. 2017 – May 2021

Research Advisor: Prof. Shaama M. Sharada

Thesis: "Investigating the Mechanisms of C–H Activation with Enzyme-Inspired Molecular Catalysts" (Awarded Best Research Paper)

RESEARCH & PROFESSIONAL EXPERIENCE

Kulik Group, MIT Department of Chemical Engineering

Graduate Research Assistant

Cambridge, MA

Jan. 2023 – Present

- Developed graph neural networks to predict organometallic bonding in transition metal complexes from ligand structure (*PNAS*), and extended the approach to dynamic coordination modes using ensemble learning (*JACS*)
- Co-led construction of BOS-TMC, the largest DFT dataset of experimentally characterized transition metal complexes in multiple charge and spin states (*JCIM*)
- Supported development and benchmarking of state-of-the-art generative transition state models (*Nat. Comput. Sci.*)
- Built ElemeNet, a unified machine learning framework for chemistry supporting E(3)-equivariant architectures with uncertainty quantification across the periodic table
- Designed novel physics-based molecular fingerprints from spectral graph theory to encode 3D information while preserving permutation and E(3) invariance
- Maintain the group HPC cluster and contribute to the open-source molSimplify structure generation package

IBM Research

Algorithms and Applications Intern

Yorktown Heights, NY

Jun. 2026 – Aug. 2026

- Developed wavefunction embedding schemes for simulating large molecular systems on quantum computers
- Benchmarked classical (CCSD, FCI) and quantum algorithms (SQD) on biologically relevant host-guest binding
- Investigated fine-tuning generative models for antibody structure prediction

SpaceX

Operations Engineer (Full-time)

Hawthorne, CA

Jun. 2021 – Aug. 2022

- Led cross-functional cost-reduction campaigns to optimize Falcon 9 production, saving over \$450K annually

ExxonMobil Research & Technology Development

Process Engineering R&D Intern

Houston, TX

Jun. 2020 – Aug. 2020

- Developed computational tool for automated heat-exchanger design, reducing engineering labor hours by 10%

Sharada Group, USC Department of Chemical Engineering and Materials Science

Undergraduate Researcher

Los Angeles, CA

Jan. 2019 – May 2021

- Performed DFT mechanistic studies of C–H hydroxylation by enzyme-inspired copper–oxygen complexes
- Published in *Catal. Sci. Technol.* and recognized with departmental awards for best research paper and presentation

TECHNICAL SKILLS

Machine Learning: geometric deep learning (GNNs, E3NN), generative models (diffusion, flow-based), uncertainty quantification, active learning, Bayesian modeling and inference

Quantum Chemistry: density functional theory (TeraChem, ORCA, Psi4, PySCF), post-Hartree–Fock methods, transition-state search, quantum computing and quantum algorithms

Scientific Computing: Python, PyTorch, PyTorch Geometric, HPC cluster administration, multi-GPU training

PUBLICATIONS

8 peer-reviewed publications and 3 preprints (first or co-first author on 7) in venues including PNAS, JACS, and Nature Computational Science. | * denotes equal contribution.

Peer-Reviewed

1. A. G. Garrison*, J. W. Toney*, T. Nikolaeva, R. G. St. Michel, C. J. Stein, H. J. Kulik, “The BOS-TMC Dataset: DFT Properties of 159k Experimentally Characterized Transition Metal Complexes Spanning Multiple Charge and Spin States”, *J. Chem. Inf. Model.* **2026**, [10.1021/acs.jcim.6c01792](https://doi.org/10.1021/acs.jcim.6c01792).
2. S. Darouich, J. W. Toney, W. Luo, J. Kästner, M. Niepert, H. J. Kulik, “Robust Generative Transition State Models for Unseen Chemistry”, *Nat. Comput. Sci.* **2026**, [10.1038/s43588-026-01034-5](https://doi.org/10.1038/s43588-026-01034-5).
3. G. G. Terrones, R. G. St. Michel, J. W. Toney, A. K. Ball, Y. Wang, A. G. Garrison, A. Nandy, R. Meyer, F. Edholm, C. Oh, S. G. Pujet, D. B. K. Chu, D. Muhammetgulyyev, H. J. Kulik, “molSimplify 2.0: Improved Structure Generation for Automating Discovery in Inorganic Molecular and Reticular Chemistry”, *J. Chem. Inf. Model.* **2026**, [10.1021/acs.jcim.5c02733](https://doi.org/10.1021/acs.jcim.5c02733).
4. J. W. Toney, R. G. St. Michel, A. G. Garrison, I. Kevlishvili, H. J. Kulik, “Identifying Dynamic Metal–Ligand Coordination Modes with Ensemble Learning”, *J. Am. Chem. Soc.* **2025**, [10.1021/jacs.5c17169](https://doi.org/10.1021/jacs.5c17169).
5. J. W. Toney, R. G. St. Michel, A. G. Garrison, I. Kevlishvili, H. J. Kulik, “Graph Neural Networks for Predicting Metal–Ligand Coordination of Transition Metal Complexes”, *Proc. Natl. Acad. Sci.* **2025**, [10.1073/pnas.2415658122](https://doi.org/10.1073/pnas.2415658122).
6. J. W. Toney, A. G. Garrison, W. Luo, R. G. St. Michel, S. Mukhopadhyay, H. J. Kulik, “Exploring Beyond Experiment: Generating High-Quality Datasets of Transition Metal Complexes with Quantum Chemistry and Machine Learning”, *Curr. Opin. Chem. Eng.* **2025**, [10.1016/j.coche.2025.101189](https://doi.org/10.1016/j.coche.2025.101189).
7. I. Kevlishvili, R. G. St. Michel, A. G. Garrison, J. W. Toney, H. Adamji, H.-J. Jia, Y. Román-Leshkov, H. J. Kulik, “Leveraging Natural Language Processing to Curate the tmCAT, tmPHOTO, tmBIO, and tmSCO Datasets of Functional Transition Metal Complexes”, *Faraday Discuss.* **2024**, [10.1039/D4FD00087K](https://doi.org/10.1039/D4FD00087K).
8. Z. Lan*, J. W. Toney*, S. M. Sharada, “A Computational Mechanistic Study of C–H Hydroxylation with Mononuclear Copper–Oxygen Complexes”, *Catal. Sci. Technol.* **2023**, [10.1039/D2CY01128J](https://doi.org/10.1039/D2CY01128J).

Preprints and Manuscripts Under Review

- P1. J. W. Toney, A. Y. Farnood, S. Darouich, H. J. Kulik, “Physics-Based Molecular Fingerprints from Spectral Graph Theory Provide Efficient Geometry-Aware Measures of Chemical Similarity”, *arXiv* **2026**, [10.48550/arXiv.2608.05336](https://arxiv.org/abs/10.48550/arXiv.2608.05336).
- P2. J. W. Toney*, S. Darouich*, Y. Wang, A. G. Garrison, J. Kästner, H. J. Kulik, “ElemeNet: Multiscale Molecular Machine Learning with Uncertainty Quantification Across the Periodic Table”, *arXiv* **2026**, [10.48550/arXiv.2606.30961](https://arxiv.org/abs/10.48550/arXiv.2606.30961).
- P3. Y. Liu, J. W. Toney, J. M. Cavanagh, K. Sun, A. L. Smith, C. Yuan, R. G. St. Michel, P. A. Graggs, D. Toste, H. J. Kulik, T. Head-Gordon, “Exploring Transition Metal Complexes with Large Language Models”, *ChemRxiv* **2025**, [10.26434/chemrxiv-2025-hm3zb](https://doi.org/10.26434/chemrxiv-2025-hm3zb).

HONORS & AWARDS

Entalpic AI for Science Fellow, Entalpic	2026
Leslye Miller Fraser and Darryl M. Fraser Fellow, MIT School of Engineering	2025
MathWorks Engineering Fellow, MIT Department of Chemical Engineering	2024
Alfred P. Sloan UCEM Fellow, MIT School of Engineering	2022
Valedictorian, USC School of Engineering	2021
University Trustees Award (perfect GPA), USC	2021
National Academy of Engineering Grand Challenges Scholar, USC School of Engineering	2021
Outstanding Chemical Engineering Student Award, USC Department of Chemical Engineering	2021
Best Research Paper and Best Research Presentation, USC Department of Chemical Engineering	2021
Presidential Scholar, USC	2017
Latino Alumni Association Scholar, USC	2017

PRESENTATIONS

1. International Conference on Quantum Chemistry, Berkeley, CA. “Machine Learning Prediction of Structural and Electronic Properties of Transition Metal Complexes”, poster. June 2026.
2. LeMaterial Reading Group, Virtual. “Exploring Structure, Properties, and Reactivity in Transition Metal Complexes with Deep Learning and Quantum Chemistry”, **invited talk**. February 2026.
3. American Institute of Chemical Engineers (AIChE), Boston, MA. “Graph Neural Networks for Predicting Metal–Ligand Coordination in Transition Metal Complexes”, talk in *Data Science and ML Approaches to Catalysis*. November 2025.
4. World Association of Theoretical and Computational Chemists (WATOC), Oslo, Norway. “Graph Neural Networks for Predicting Metal–Ligand Coordination in Transition Metal Complexes”, talk in *Chemical Structure and Reactivity*. June 2025.
5. CECAM Frontiers of Computational Reaction Prediction, Chicago, IL. “Graph Neural Networks for Predicting Metal–Ligand Coordination in Transition Metal Complexes”, poster. July 2024.
6. USC ChE & Materials Science Research Symposium, Los Angeles, CA. “Investigating the Mechanisms of C–H Activation with Enzyme-Inspired Molecular Catalysts”, recorded talk. February 2021. **Awarded Best Research Presentation.**
7. AIChE Annual Student Conference, San Francisco, CA. “Computational Analysis of C–H Hydroxylation in Bio-inspired Monometallic Complexes”, poster. November 2020.
8. USC ChE & Materials Science Research Symposium, Los Angeles, CA. “Computational Analysis of C–H Hydroxylation in Bio-inspired Monometallic Complexes”, poster. March 2020.
9. AIChE Annual Student Conference, Orlando, FL. “Computational Analysis of C–H Hydroxylation in Bio-inspired Monometallic Complexes”, poster. November 2019.

TEACHING

5.698/10.637 Computational Chemistry (MIT): Grader, graduate elective	Fall 2025
10.34 Numerical Methods (MIT): Graduate Teaching Fellow, graduate core	Fall 2024
<i>Instructor rating 6.5/7.</i>	
10.65 Chemical Reactor Engineering (MIT): Grader, graduate core	Spring 2024
10.40 Chemical Engineering Thermodynamics (MIT): Grader, graduate core	Fall 2023

MENTORSHIP & SERVICE

Undergraduate Researchers Mentored

1. **David Sauer** (Chemistry, Harvard University): Reparameterization of electronic structure methods for transition metal complexes.
2. **Zheming Zhang** (Chemical Engineering, MIT): Benchmarking deep learning architectures for electronic property prediction in variable charge and spin states.
3. **Lauren Wright** (Chemical Engineering, Louisiana State University): Designing new hydroformylation catalysts through mechanistic study and data science. Presented at AIChE Annual Student Conference, Boston, MA, November 2025; **Awarded Third Place Poster.**
4. **Sebastian Pujet** (Chemical Engineering, Washington University in St. Louis): Optimizing mechanochemical properties of transition metal complexes with active learning . Contributed to molSimplify 2.0 (*JCIM* 2026).

Leadership

President , USC AIChE Student Chapter	Apr. 2020 – May 2021
President , USC Latino Alumni Association Scholar Committee	Jan. 2019 – May 2021